

MAP READING COURSE MANUAL



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MILITARY TRAINING COMMAND

Kali Special Operations Legion, Headquarters

Foreword by the Commander

1. PURPOSE

Document KSOL-MR 5-1A (Map Reading Course Manual) establishes the fundamental knowledge and procedures required for personnel to read, interpret, and utilise maps for navigation and positional awareness within KSOL operations.)

2. SCOPE

This document covers map grids and grid references, grid-reference precision, MGRS, distance measurement and estimation, types of North, compass directions, and bearings. It provides personnel with the foundational skills required to determine and communicate positions, distances, and directions using a map.

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APPENDIX 'A'

Map Reading Fundamentals

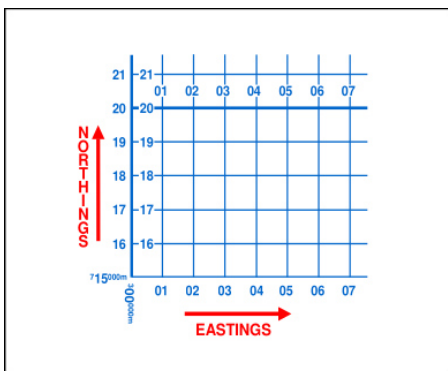
AA01. Map Grids

A map grid helps to find a location/position on a map using a **Grid Reference**.

Grids are made of **two sets parallel lines** on map, which forms **grid squares**. Set of lines moving up & down are called **Northings**, corresponding to **y-axis** in a cartesian plane. And set of lines moving left to right are called **Eastings**, corresponding to **x-axis** in a cartesian plane.

AA02. Grid Reference

A grid reference indicates a location in terms of series of vertical and horizontal lines identified by numbers/letters.



NOTE: Always read the EASTINGS FIRST followed by NORTHINGS.

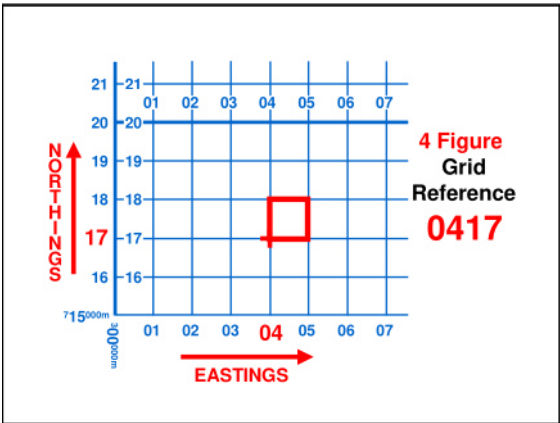
The accuracy of grid reference depends on the figures. Grid references can be of **4, 6, 8, and 10** figures.

** in order of least to most precise*

No. of Figures	Accuracy Upto
4	1km × 1km
6	100m × 100m
8	10m × 10m
10	1m × 1m

NOTE: ONLY 4-figure, 6-figure, 8-figure are RELEVANT FOR BRM5.

Example (using 4-figure reference)



Here to read/refer to the marked grid square, for **Easting** we consider the closest **left-most** line. And for **Northing**, we consider the closest bottom-most line.

Notice how if these two left-most and bottom-most lines together

mimic the cartesian y-axis and x-axis.

AA03. MGRS (Military Grid Reference System)

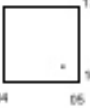
IRRELEVANT FOR BRM5*

This is the NATO standard grid referencing system which is used on ALL operational mapping and navigation systems. IT CANNOT BE USED ON BRITISH OR IRISH MAPS.

MGRS consists of THREE PARTS – (1) and (2) which you already know:-

- (1) 047 162 - 6 Figure Grid Reference
- (2) NS –100 km (100,000 metres) Grid Square Letters (found on the map or in the map margin box as shown below)
- (3) 32R – Grid Zone Designator-GZD-(found on the map or in the map margin box as shown below)

*MGRS will be WRONG if the DATUM is WRONG. Check that you are using the correct DATUM as printed on your map and used in your GPS.**

SAMPLE 1,000 METRE GRID SQUARE (1) 	100 METRE REFERENCE 1. Read the large numbers labeling the VERTICAL grid line from left of point and estimate tenths (100metres) from grid line to point. 04 7 2 Read the large numbers labelling the HORIZONTAL grid line below point and estimate tenths (100metres) from grid line to point. 16 2 Example: 047 162				
100,000 METRE SQUARE IDENTIFICATION (2) <table border="1" data-bbox="330 1125 442 1204"> <tr> <td>NT</td> <td>PT</td> </tr> <tr> <td>NS</td> <td>PS</td> </tr> </table> 3200 888	NT	PT	NS	PS	WHEN REPORTING ACROSS A 100,000 METRE LINE, PREFIX THE 100,000 METRE SQUARE IDENTIFICATION IN WHICH THE POINT LIES Example: NS 047 162
NT	PT				
NS	PS				
GRID ZONE DESIGNATOR (3) 32 R	WHEN REPORTING OUTSIDE THE GRID ZONE DESIGNATOR AREA, PREFIX THE GRID ZONE DESIGNATOR Example: 32R NS 047162				

AA04. Types of Measuring Distances

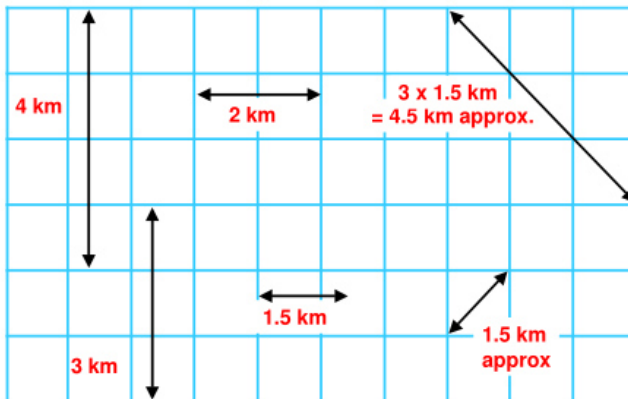
There are two ways of measuring distance on a map:

Direct (Straight line) and **Indirect** (along Roads/Routes).

For our faction we will use Direct measurement.

AA05. Estimating Direct Distances Using Map Grid

Grid squares are **usually 1km** (kilometers) apart and about **1.5km** (a little less than 1.5km) diagonally.



Linear distance = No. of grids × 1km

Diagonal distance = No. of diagonal grids × 1.5km

AA06. Types of North

AA06 A. Magnetic North

The compass needle points to Magnetic North.

AA06 B. Grid North

The north-south (vertical) grid lines point to grid north at the top of the map. Grid north is not a place, just the direction of all the grid lines.

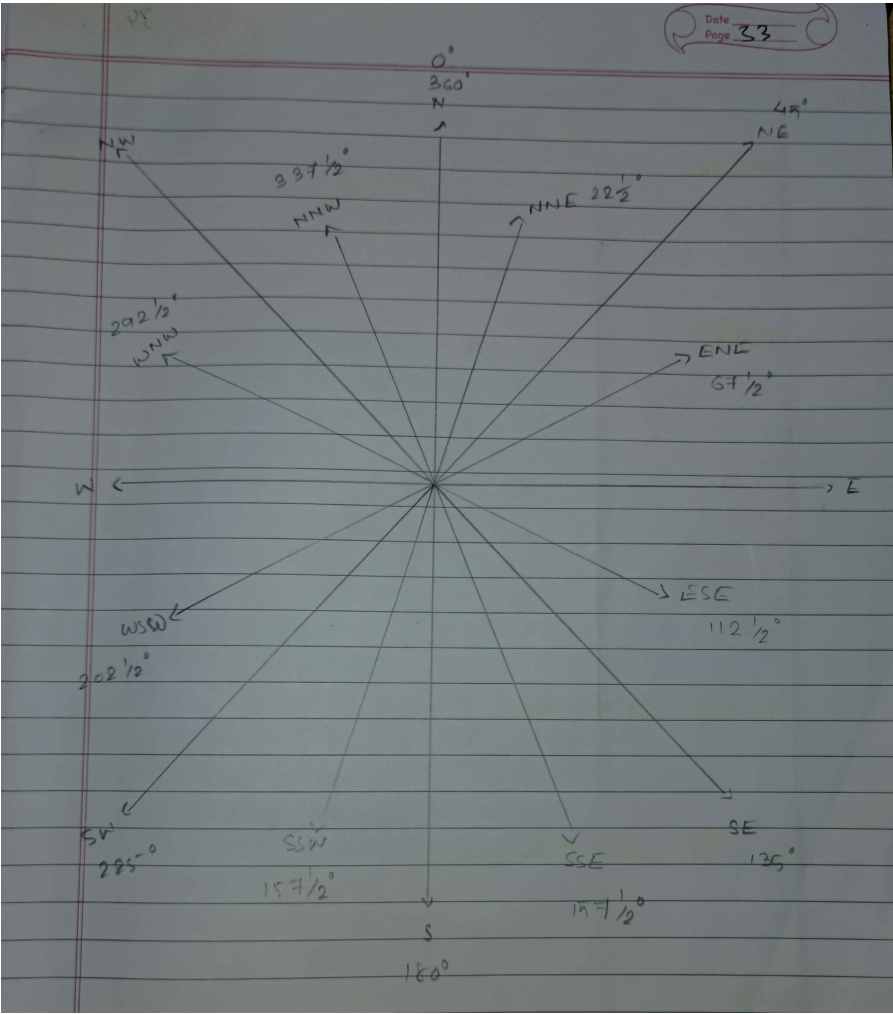
AA06 C. True North

This is the direction to the North Pole.

AA07. Directions & Types

Type	Directions	Degrees (in deg)
Cardinal	N,S,E,W	0°, 90°, 180°, 270°
Intercardinal/Ordinal	NE,SE,NW,SW	45°, 135°, 225°, 315°
Secondary Intercardinal	NNE, ENE, ESE, SSE, SSW, WSW, WNW, NNW	In intervals of 22.5°

Look at the image on the next page for consolidated data.



AA08. Bearings

A **bearing** is a numerical representation of direction measured **clockwise from North**. Bearings are expressed in degrees from **000° to 360°**, with North serving as the reference direction.

“MEN APART, EVERY MAN AN EMPEROR”

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